

Smooth Object Pose Trajectory Estimation

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1 Problem Setting

The objective was to estimate a pose for an object template in every video frame. For template vertex $\mathbf{v}_0^{(i)}$, I represented its position in the camera frame as

$$\mathbf{v}_t^{(i)} = s\mathbf{R}_t\mathbf{v}_0^{(i)} + \mathbf{t}_t, \quad \mathbf{R}_t \in SO(3), \quad \mathbf{t}_t \in R^3, \quad s > 0.$$

The available evidence included object point cloud, object mask, camera intrinsics, and a template mesh. The main difficulties were human occlusion, isolated reconstruction outliers, and orientation ambiguity for symmetric or nearly symmetric objects. I first tested the methods on manually selected frame intervals before producing each complete trajectory. The intervals were categorized into three: **clear**, where the object was mostly visible and observations were reliable; **motion**, where the object underwent substantial movement; and **occlusion**, where the object was partially hidden by the person. Such categories elucidate the dominant condition and are not necessarily mutually exclusive, as a clear interval may still contain some motion, for example.

I evaluated alignment using silhouette intersection over union (IoU), symmetric unsquared Chamfer distance, and observation-to-mesh distance. Temporal behavior was measured using geodesic rotation change and Euclidean translation change between consecutive frames. Overlay videos were also inspected manually as a low geometric residual can still correspond to a visually plausible but incorrect orientation.

2 Approaches Explored

The experiments progressed from simple geometric alignment to methods with increasingly strong temporal information. Table 1 summarizes why each method was tested and the principal outcome.

Table 1: Comparison of the evaluated approaches.

Approach	Rationale and Method	Advantages Observed	Disadvantages Observed
0: Centroid/PCA	Align the template and observation centroids, optionally aligning their principal axes.	PCA greatly improved the initial suitcase, umbrella, and chair alignment and provided a useful orientation initializer.	PCA-axis signs were ambiguous, producing an inverted suitcase and repeated umbrella flips. Ball orientation was unobservable, and method followed a corrupted ball cloud at certain frame.
1: Independent ICP	Refine each frame with point-to-point ICP. The multi-start variant evaluates 24 valid combinations of PCA axis order and direction.	Multi-start ICP escaped several PCA local minima and recovered plausible upright suitcase, umbrella, and chair poses with lower Chamfer distance.	Frames remained independent, so symmetry-equivalent branches could change between frames. Umbrella rotation and the ball translation outlier remained.

Table 1 - continued

Approach	Rationale and Method	Advantages Observed	Disadvantages Observed
2: Temporal filtering	Smooth Approach 1 translation with a Savitzky-Golay filter and rotation with a Markley quaternion mean.	Larger windows strongly reduced chair wobble, umbrella flips, and numerical frame-to-frame variation.	Filtering could not correct a biased pose. It spread the isolated ball outlier into neighbouring frames and could suppress genuine motion.
3: Sequential ICP	Initialize each frame from the preceding pose. The robust variant adds a distance gate and trimmed observation-to-template correspondences.	Removed the catastrophic ball displacement; large suitcase and umbrella orientation jumps without distributing an error over a long window.	The trajectory remained sensitive to its boundary orientation and tracking direction. Trimming could discard informative thin or partially visible geometry.
4: Temporal candidate path	Generate several ICP candidates per frame and use dynamic programming to minimize data and transition costs over the interval.	Removed large suitcase candidate switches and substantially reduced the defined combined cost of fitting each frame and avoiding sudden changes.	It could only select existing candidates, so it could not fix suitcase depth bias, reject the ball’s sole corrupted candidate, or fully stabilize the umbrella.
5: Joint optimization	Jointly refine geometry with velocity, acceleration, and optional silhouette-contour terms.	Reduced average frame-to-frame motion and explicitly demonstrated the trade-off between observation fit and temporal smoothness.	It generally increased Chamfer distance, while the mask term gave only small IoU gains. Its additional complexity did not provide a consistent overall improvement.

These experiments showed that optimizing each frame independently was not sufficient: a pose could have a low residual while using a different geometrically similar orientation. Conversely, strong post-processing could make an erroneous trajectory smoothly wrong. Sequential initialization gave the best general balance because it prevented isolated failures while retaining more genuine motion than a large smoothing window.

3 Final Method

The submitted trajectories use one uniform method, referred to as **Approach 3S**. I first initialized the boundary frame using the best multi-start ICP pose and then tracked forward through the complete sequence. Each frame started from the preceding estimate. Robust sequential ICP used matches from each observed point to its nearest template point, a maximum-distance gate, and trimming to reduce the influence of outliers and template regions that were absent from a partial observation. The ball used a fixed identity rotation because rotating a spherical template looks the same under any rotation.

I then applied the same conservative five-frame smoothing configuration to all four sequences. Translation was filtered with a second-order Savitzky-Golay filter, and rotation was averaged using a triangularly weighted Markley quaternion mean so that every result remained in $SO(3)$. The scale was fixed at $s = 1$ and frame indices were preserved. This final stage targeted small high-frequency wobble rather than attempting to repair large fitting failures.

4 Results

Table 2 reports complete-sequence results for Approach 3S. Chamfer and translation changes are reported in centimeters, while rotation change is reported in degrees.

Table 2: Full-trajectory evaluation of the submitted results.

Sequence	Object	Mean IoU	Mean Chamfer	Mean ΔR	Mean Δt
01_01	Suitcase	0.782	3.41 cm	1.31°	3.07 cm
01_03	Ball	0.856	2.29 cm	0.00°	1.85 cm
01_04	Umbrella	0.688	2.91 cm	1.03°	1.56 cm
01_07	Chair	0.445	3.88 cm	1.86°	2.78 cm

No submitted trajectory contained a frame-to-frame rotation change above 60°. The ball gave the strongest quantitative result; its translation was stable and its deliberately fixed rotation avoided meaningless symmetry-equivalent changes. The suitcase remained upright and continuous, although its projected footprint was sometimes smaller than the observed mask. The umbrella canopy tracked reliably, but the handle alignment was less certain. The chair remained reasonably aligned through much of the sequence, including periods of heavy occlusion, but gradually drifted near the end: its estimated pose rotated to the right and its projected size increased as the person moved away and the visible chair geometry changed. Because this error accumulated through several small pose changes rather than a single large jump, it was not captured by the 60° frame-to-frame threshold, demonstrating that the absence of abrupt discontinuities does not guarantee an accurate trajectory.

5 Weaknesses and Limitations

The chair produced the weakest result according to silhouette IoU and also developed gradual pose drift. When the person sat on the chair, much of it was hidden and the supplied mask contained only its visible regions. The evaluation rendered the complete chair template, so hidden but correctly positioned surfaces were counted as predicted-only regions. As the person moved away near the end of the video, the estimated chair gradually rotated to the right and its projected size increased. Since the scale remained fixed at $s = 1$, this apparent growth was most likely caused by errors in estimated depth and rotation. These errors accumulated through several small frame-to-frame changes, so they were not detected by the 60° rotation-jump threshold. This shows that a trajectory can be free of sudden jumps while still drifting away from the correct pose over time.

The umbrella has a related observability problem. Its broad, approximately symmetric canopy dominates both the point cloud and silhouette, whereas the thin handle is often missing or weakly represented. Several handle orientations can therefore obtain similar fitting costs. The suitcase’s smaller projected footprint is another systematic limitation. Because scale was fixed, this is most likely caused by depth and translation bias in the partial reconstructed geometry rather than an explicitly shrinking scale. More generally, sequential tracking can maintain

a consistent orientation, but it cannot prove that the initially chosen orientation is correct or recover information absent under severe occlusion.

6 Conclusion

Among the tested approaches, the most reliable method was to use the pose from the previous frame as the starting point for the next frame and to ignore poorly matching point pairs. This prevented the isolated ball reconstruction error from moving the pose to the wrong location and reduced sudden orientation changes in the suitcase and umbrella trajectories. A short smoothing step then reduced the remaining frame-to-frame wobble. However, the method still struggled when important parts of an object were hidden or missing from the observations, as seen in the umbrella handle and the gradual chair drift. Overall, the final method produced smoother and more consistent trajectories, but it could not fully recover pose information that was not visible in the input data.